

04 — Gate 12 Screening-Tier Benchmark Report

SEION-KGR v26 MAX (NOT a confirmatory report)

Campaign ID: gate12-closeout-2026-08-01
Canonical commit: 6af3c35271ae2ffab41ecba2aad098d1988fdc0c
Branch: campaign/gate12-closeout
Date: 2026-08-01

Purpose: report the Tier 0 numbers in full, with the confound stated before any interpretation,
not after.

Contents

1	Datasets	1
2	Base-expert selection	1
3	A0 (TuckER alone, full dataset) vs. A3 (TuckER+path+seion, 5,000-triple sub-sample)	2
4	The A0/A3 comparison is confounded — stated before interpretation	2
5	Head/tail breakdown	2
6	Compute	2
7	Statistical treatment	3

Read this first

This is a 2-seed screening pass, not a Gate 12 confirmatory result. The mandate’s own protocol (preregistration.md §5/§8) requires 5 seeds, full dataset scale, the full 13-point ablation ladder, and statistical testing with multiple-comparison correction before any number here could support a confirmatory claim. None of that was executed. Every number below is labeled **SCREENING (reduced)**.

1 Datasets

Dataset	Split	SHA-256 (first 16 hex)
FB15K-237	train	6e4c2782169af21e
FB15K-237	valid	cf6309010852f6a8
FB15K-237	test	5711cf41623ceb4e
WN18RR	train	038612e783c215ee
WN18RR	valid	453ce7202afa5809
WN18RR	test	0383bceaaa1096cf

Full hashes: campaigns/gate12/preregistration.md §2.

2 Base-expert selection

1-epoch screening, FB15K-237, dim=64, eval_max_queries=200, seed=1, real GPU:

Base expert	Valid MRR	Test MRR
TuckER (selected)	0.2534	0.2526
ComplEx	0.2210	0.2459
DistMult	0.2195	0.2475
CP	0.2092	0.2140

3 A0 (TuckER alone, full dataset) vs. A3 (TuckER+path+seion, 5,000-triple subsample)

Config	Dataset	Seed	Test MRR	Valid MRR	Hits@1	Hits@10	Wall (s)
A0	FB15K-237	1	0.2795	0.2912	0.1917	0.4450	78.2
A0	FB15K-237	2	0.2774	0.2895	0.1867	0.4550	84.9
A0	WN18RR	1	0.4414	0.4140	0.4233	0.4767	22.3
A0	WN18RR	2	0.4235	0.4073	0.3933	0.4700	24.3
A3	FB15K-237	1	0.0095	0.0119	0.0050	0.0250	217.2
A3	FB15K-237	2	0.0229	0.0090	0.0200	0.0300	214.6
A3	WN18RR	1	0.0018	0.0034	0.0000	0.0050	53.4
A3	WN18RR	2	0.0030	0.0138	0.0000	0.0150	53.4

4 The A0/A3 comparison is confounded — stated before interpretation

A3 trained on a 5,000-triple subsample (1.8% of FB15K-237’s, 5.8% of WN18RR’s full training split) for 3 epochs; A0 trained on the full split for 5 epochs. This deviation was logged in `preregistration.md` §11 *before* any A3 result was generated (a live timing probe showed the full-scale A3 configuration did not complete one epoch in over 8 minutes; see `09_negative_results.pdf`). **The large MRR gap between A0 and A3 must not be read as evidence about the path/seionic branches** — a same-data-scale, same-epoch-budget comparison was not executed.

5 Head/tail breakdown

Config	Dataset (seed-1)	Head MRR	Tail MRR
A0	FB15K-237	0.1929	0.3662
A0	WN18RR	0.4209	0.4620
A3	FB15K-237	0.0015	0.0175
A3	WN18RR	0.0021	0.0016

A0’s head/tail ratio (≈ 0.53 FB15K-237, ≈ 0.91 WN18RR) is within the reasonable range the preregistration’s finalist-selection rule declared ($[0.3, 3.0]$); no severe collapse observed for the base expert alone at this screening scale.

6 Compute

All A0 runs used the real RTX PRO 5000 Blackwell (GPU faster for pure-GEMM base-expert scoring). All A3 runs used CPU (GPU was measurably slower for the path reasoner’s per-sample workload during the timing probe — see 09). No VRAM/host-RAM profiling was collected in this campaign (a real gap, listed in 09).

7 Statistical treatment

None applied. 2 seeds is reported per-seed only (above), never as $\text{mean} \pm \text{std}$, per `preregistration.md` §9's explicit instruction that this would misrepresent a 2-point sample as a distributional estimate.